

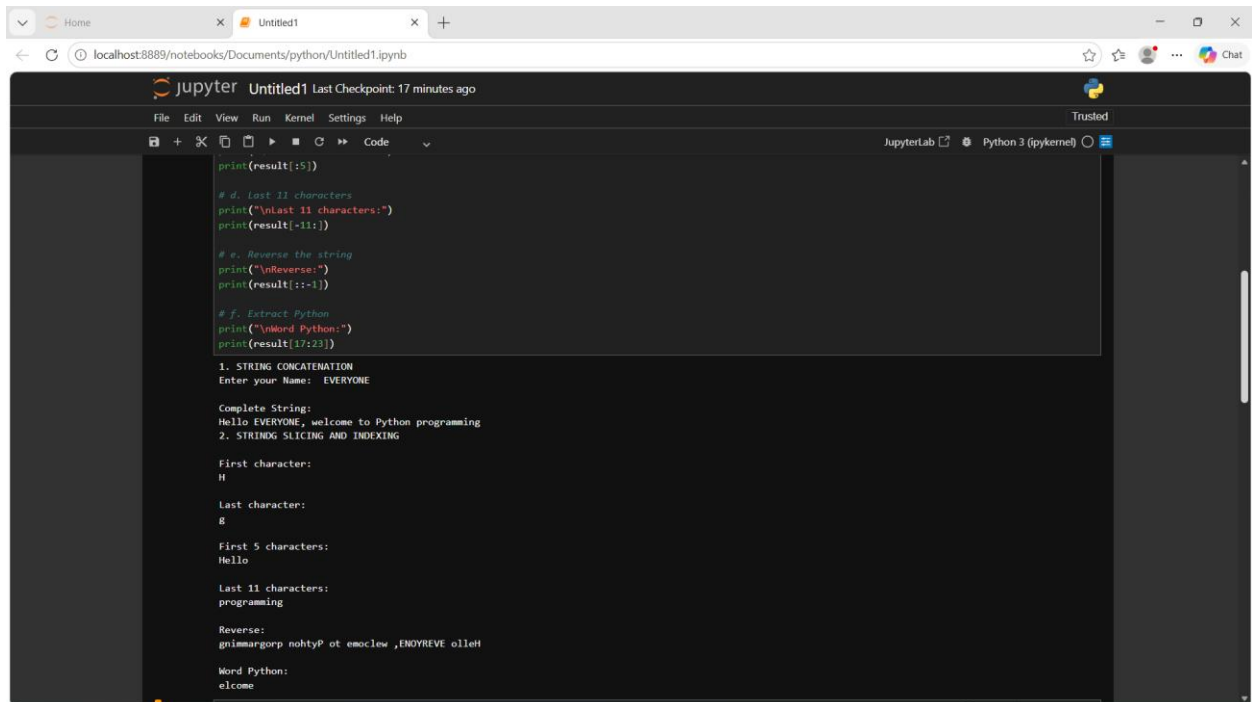
PYTHON – ASSIGNMENT 1

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DATA STRUCTURES – STRINGS & TUPLES

SCREENSHOTS

Task 1 – String Concatenation



The screenshot shows a JupyterLab interface with a notebook titled 'Untitled1'. The code in the notebook is as follows:

```
print(result[:5])

# d. Last 11 characters
print("\nLast 11 characters:")
print(result[-11:])

# e. Reverse the string
print("\nReverse:")
print(result[::-1])

# f. Extract Python
print("\nWord Python:")
print(result[17:23])
```

The output of the code is displayed below the code cell:

```
1. STRING CONCATENATION
Enter your Name:  EVERYONE

Complete String:
Hello EVERYONE, welcome to Python programming
2. STRING SLICING AND INDEXING

First character:
H

Last character:
g

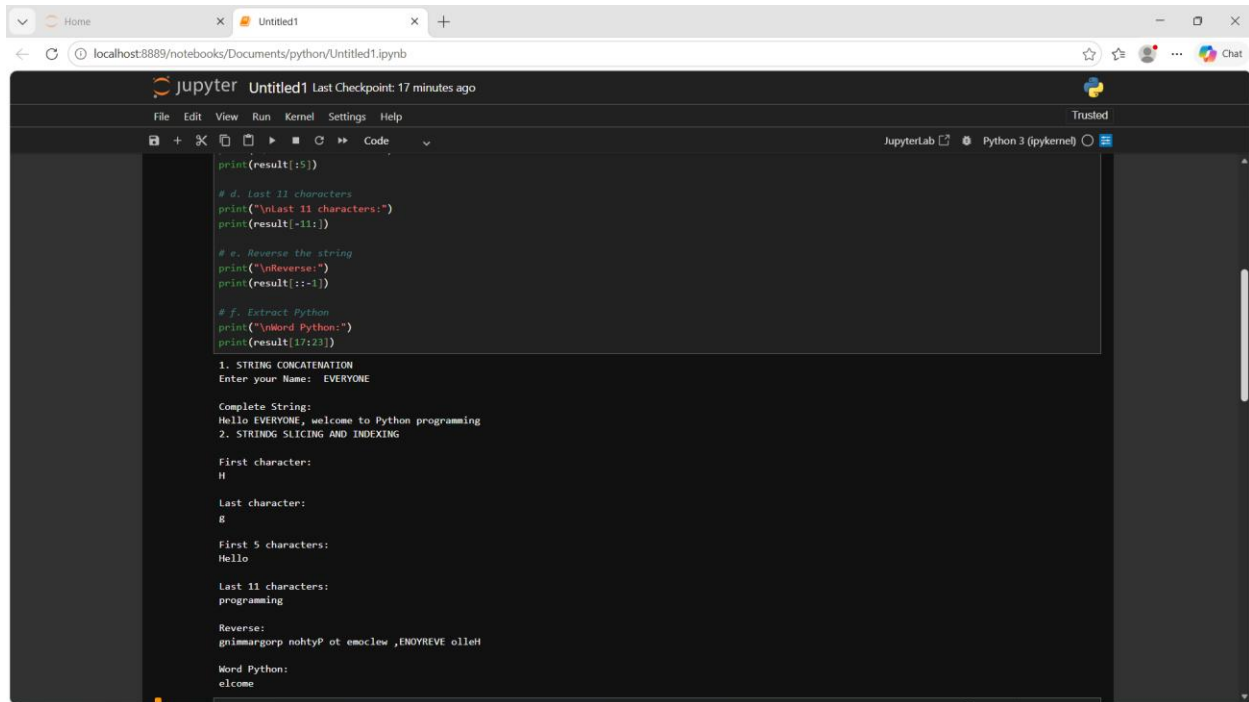
First 5 characters:
Hello

Last 11 characters:
programming

Reverse:
gnimargorp nohtyP ot eoclew ,ENOYREVE olleH

Word Python:
elcome
```

Task 2 – String Slicing and Indexing



The screenshot shows a JupyterLab notebook titled 'Untitled1' with a Python 3 kernel. The code in the notebook demonstrates various string slicing and indexing techniques on the string 'Hello EVERYONE, welcome to Python programming'.

```
print(result[:5])

# d. Last 11 characters
print("\nLast 11 characters:")
print(result[-11:])

# e. Reverse the string
print("\nReverse:")
print(result[::-1])

# f. Extract Python
print("\nWord Python:")
print(result[17:23])
```

1. STRING CONCATENATION
Enter your Name: EVERYONE

Complete String:
Hello EVERYONE, welcome to Python programming

2. STRING SLICING AND INDEXING

First character:
H

Last character:
E

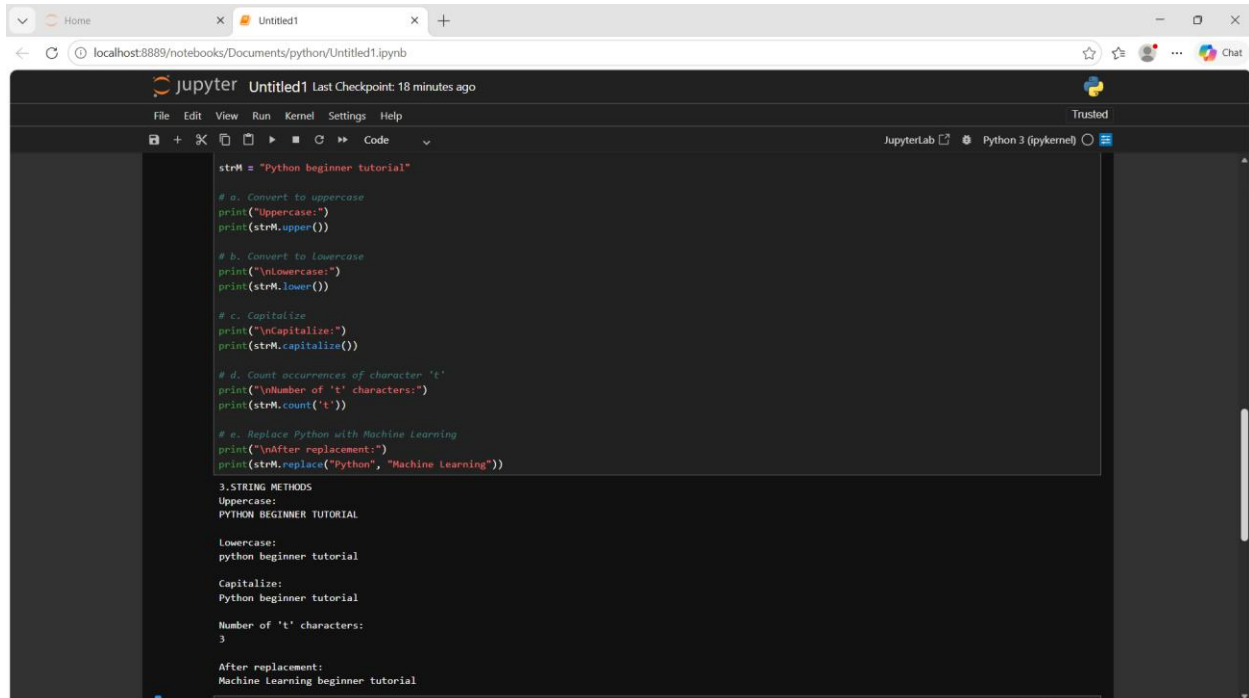
First 5 characters:
Hello

Last 11 characters:
programming

Reverse:
gnimargorp nohtyP ot emoclew ,ENOYREVE olleH

Word Python:
elcome

Task 3 – String Methods



The screenshot shows a JupyterLab notebook titled 'Untitled1' with a Python 3 kernel. The code in the notebook demonstrates various string methods on the string 'Python beginner tutorial'.

```
strM = "Python beginner tutorial"

# a. Convert to uppercase
print("\nUppercase:")
print(strM.upper())

# b. Convert to lowercase
print("\nLowercase:")
print(strM.lower())

# c. Capitalize
print("\nCapitalize:")
print(strM.capitalize())

# d. Count occurrences of character 't'
print("\nNumber of 't' characters:")
print(strM.count('t'))

# e. Replace Python with Machine Learning
print("\nAfter replacement:")
print(strM.replace("Python", "Machine Learning"))
```

3. STRING METHODS

Uppercase:
PYTHON BEGINNER TUTORIAL

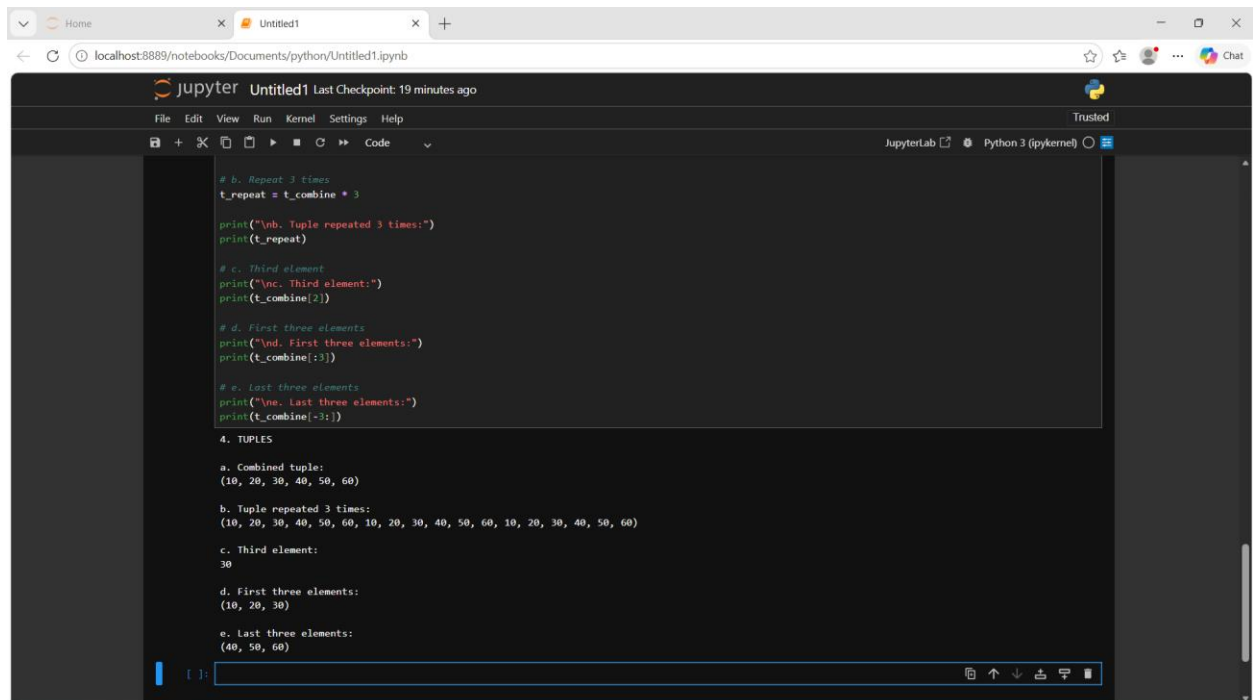
Lowercase:
python beginner tutorial

Capitalize:
Python beginner tutorial

Number of 't' characters:
3

After replacement:
Machine Learning beginner tutorial

Task 4 – Tuples



The screenshot shows a JupyterLab interface with a notebook titled 'Untitled1'. The code in the notebook is as follows:

```
# b. Repeat 3 times:
t_repeat = t_combine * 3

print("\nb. Tuple repeated 3 times:")
print(t_repeat)

# c. Third element
print("\nc. Third element:")
print(t_combine[2])

# d. First three elements
print("\nd. First three elements:")
print(t_combine[:3])

# e. Last three elements
print("\ne. Last three elements:")
print(t_combine[-3:])
```

The output of the code is displayed below the code cells:

```
4. TUPLES

a. Combined tuple:
(10, 20, 30, 40, 50, 60)

b. Tuple repeated 3 times:
(10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)

c. Third element:
30

d. First three elements:
(10, 20, 30)

e. Last three elements:
(40, 50, 60)
```